

REVIEW ARTICLE

Intervention Effect of Child Life Specialists on Paediatric Patients: A Systematic Review of Randomized Controlled Trials

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ABSTRACT

Background: Child life specialists (CLSs) are increasingly recognized in paediatric practice, yet evidence regarding the efficacy of CLS intervention in children remains insufficient. This review aims to comprehensively evaluate the intervention effects of CLSs on paediatric patients.

Methods: Six databases (PubMed, Web of Science, Embase, Cochrane Library, CINAHL and APA PsycArticles) were systematically searched using subject terms and free terms. Randomized controlled trials (RCTs) assessing CLS effects on paediatric pain and psychological outcomes were included, with searches from database inception through 30 January 2026. Two reviewers independently screened literature and extracted data. Methodological quality was assessed using the Cochrane Risk of Bias tool. Meta-analysis was conducted using Stata/MP 18.0, calculating standardized mean difference (SMD) with 95% confidence interval (CI) via a random-effects model.

Results: The initial search yielded 1140 records, and 12 studies ($n = 1096$ participants) were ultimately included in the analysis. The included studies primarily originated from the United States and Canada. Intervention settings included inpatient units, outpatient clinics and emergency departments. The meta-analysis revealed that, compared to conditions without CLS involvement, the presence of CLSs may alleviate children's pain ($SMD = -0.28$, 95% CI -0.49 to -0.07 , $p = 0.009$), distress ($SMD = -0.26$, 95% CI -0.48 to -0.05 , $p = 0.018$) and anxiety ($SMD = -0.37$, 95% CI -0.65 to -0.01 , $p = 0.007$) during medical procedures and treatments. However, the effect on alleviating children's fear was not statistically significant ($SMD = -0.11$, 95% CI -0.32 to 0.10 , $p = 0.290$).

Conclusion: Current evidence supports the potential of CLSs to alleviate paediatric pain, psychological distress and anxiety. However, their efficacy in reducing fear requires further validation. Despite the uncertainties involved, the potential value of CLSs in clinical practice warrants broader recognition.

1 | Introduction

Children experience negative emotions such as anxiety and fear during their development, which are exacerbated when they

enter a hospital, a setting far removed from the familiarity of home. Unfamiliar medical staff, a noisy environment, painful medical procedures and the absence of family support can lead to feelings of anger, confusion, anxiety and helplessness

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Summary

- CLSs adopt a child-centred perspective and have beneficial effects on key domains of children's psychosocial experiences (specifically pain, distress and anxiety) in healthcare settings.
- A meta-analysis of 12 RCTs showed that the presence of CLSs has the potential to significantly reduce children's pain, distress and anxiety during medical procedures, whereas the effect on alleviating fear did not reach statistical significance.
- The potential value of CLSs in clinical practice warrants broader recognition and integration into paediatric care.

(Baghele et al. 2019; Dehghan et al. 2017). Research indicates that 42% of children experience significant preoperative anxiety (Moura et al. 2016). Prior to venipuncture, moderate anxiety is observed in 13% of children, while high to extremely high anxiety can reach 24.7% (Baliga et al. 2025). If these negative emotions are not effectively alleviated, they can not only hinder healthcare providers from delivering timely and efficient diagnosis and treatment, prolong wound healing and impede recovery but also increase the risk of hospital readmission, place a financial burden on families and even cause long-term psychological effects on children, leading to more intense negative emotions and pain during future medical procedures (Wittenberg B and Barnhart 2025; Abdi et al. 2025; Doupnik S et al. 2018). In the midst of busy medical work, alleviating children's pain during procedures, mitigating negative emotions and safeguarding their psychosocial health remain persistent challenges for paediatric healthcare professionals.

In recent years, hospitals and departments have made numerous attempts to improve the patient experience for children during medical visits or hospitalizations, achieving significant progress. These include distraction techniques using humanoid robots (Ali et al. 2021) or tablets (Bryl et al. 2021), therapeutic play (Dehghan et al. 2017), virtual reality (VR)-assisted therapy (Goldman R and Behboudi 2021; Bedner et al. 2021) and the introduction of the specialized skills of child life specialists (CLSs) (Schlechter et al. 2017; Fletcher et al. 2021). CLSs receive systematic education in child development and complete clinical internships and a certification examination. A CLS is a member of the multidisciplinary healthcare team. Adopting a child-centred perspective, the team collaborates to meet the needs of children, caregivers and their clients in various settings through methods such as medical play, distraction strategies and family support (Romito et al. 2021). Several studies have indicated that CLSs can improve children's psychosocial outcomes, facilitate effective communication between patients and healthcare providers, enhance the medical experience and alleviate negative emotions (Mikuluk et al. 2023; Brown et al. 2022). However, other studies suggest that services involving CLSs are no more effective than standard care in reducing children's pain, fear and anxiety (Lee-Krueger et al. 2021; Stevenson et al. 2005).

Due to variations in participant inclusion criteria, small sample sizes and inconsistent outcome measures in existing randomized controlled trials (RCTs) of CLSs in paediatric patients, there is a lack of robust evidence regarding the effectiveness of CLS services. Therefore, this study aims to conduct a comprehensive literature search and systematic review to deeply analyse the effects of CLSs on alleviating pain and improving psychosocial outcomes in paediatric patients.

2 | Methods

This study adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al. 2020) for reporting and has been registered on the PROSPERO platform.

2.1 | Search Strategy

The databases utilized in this study include PubMed, Web of Science, Embase, Cochrane Library, CINAHL and APA PsycArticles. The search period extends from the inception of each database until 30 January 2026. To avoid missing studies that employed CLS interventions but did not mention outcome measures in their titles or abstracts, this paper used population, intervention and study type as search terms. Subject terms and the free words "Child life specialist", "Child" and "Randomized Controlled Trial" were used for the search (Data S1).

2.2 | Inclusion and Exclusion Criteria for the Literature

The inclusion criteria for this study were as follows: (1) participants must be paediatric patients aged 0–18 years; (2) the intervention must involve the participation of CLSs, whereas the control group must not involve CLSs; (3) the study must be an RCT; and (4) the research site is healthcare settings.

The exclusion criteria were as follows: (1) literature for which the full text was inaccessible; (2) review articles, conference papers, dissertations, etc.; (3) scenarios that simulate hospitalization or examination.

2.3 | Study Selection and Data Extraction

All the literature was independently screened, and data were extracted by two researchers according to the predefined inclusion and exclusion criteria, followed by cross-checking. In the event of any disagreement between the two researchers, consensus was reached through discussion or by consulting a third researcher.

2.4 | Statistical Methods

Data analysis was performed using Stata/MP 18.0 software, and literature quality was evaluated using RevMan 5.4. For

continuous outcomes, standardized mean difference (SMD) with 95% confidence interval (CI) was calculated. Heterogeneity among studies—which evaluates variability in results to ensure pooled effect estimates are valid—was assessed using the chi-square test ($\alpha = 0.05$) and quantified by the I^2 statistic. I^2 values of < 25%, 25%–75% and > 75% represented low, moderate and high heterogeneity, respectively (Higgins et al. 2003). A random-effects model was applied for meta-analysis, and narrative synthesis was conducted when statistical pooling was inappropriate. Sensitivity analysis was performed by sequentially omitting individual studies.

3 | Results

A total of 1140 articles were retrieved, and after a thorough screening process, 12 articles (Schlechter et al. 2017; Lee-Krueger et al. 2021; Stevenson et al. 2005; Fletcher et al. 2023; Hinze et al. 2020; West et al. 2020; Diener et al. 2019; Dumoulin et al. 2019; Hyland et al. 2015; Heilbrunn et al. 2014; Tyson et al. 2014; Cavender et al. 2004) were ultimately included in the analysis. The literature screening process is illustrated in Figure 1.

Among the 12 articles included, seven (Schlechter et al. 2017; Stevenson et al. 2005; Hinze et al. 2020; Diener et al. 2019; Heilbrunn et al. 2014; Tyson et al. 2014; Cavender et al. 2004) were published in the United States, with publication dates ranging from 2004 to 2025. Canada was the next most common country of origin with four reports (Lee-Krueger et al. 2021; Fletcher et al. 2023; West et al. 2020; Dumoulin et al. 2019). The sample size ranged from 30 to 149 paediatric patients, totalling 1096 cases, as detailed in Table 1. The studies covered a variety of procedures, including dental restoration, preoperative care, venipuncture, fracture management, burn dressing changes and imaging examinations as detailed in Table 2.

The quality assessment of the literature included in this study was independently evaluated by two researchers following the Cochrane Handbook for Systematic Reviews (version 5.1.0) (Green and Higgins 2011). A consensus was reached to establish the final quality assessment of the literature. In cases of discrepancies, a third researcher was consulted. Among the included studies, only one study (Diener et al. 2019) did not report the generation of random sequences. Seven studies (Lee-Krueger et al. 2021; Stevenson et al. 2005; West et al. 2020;

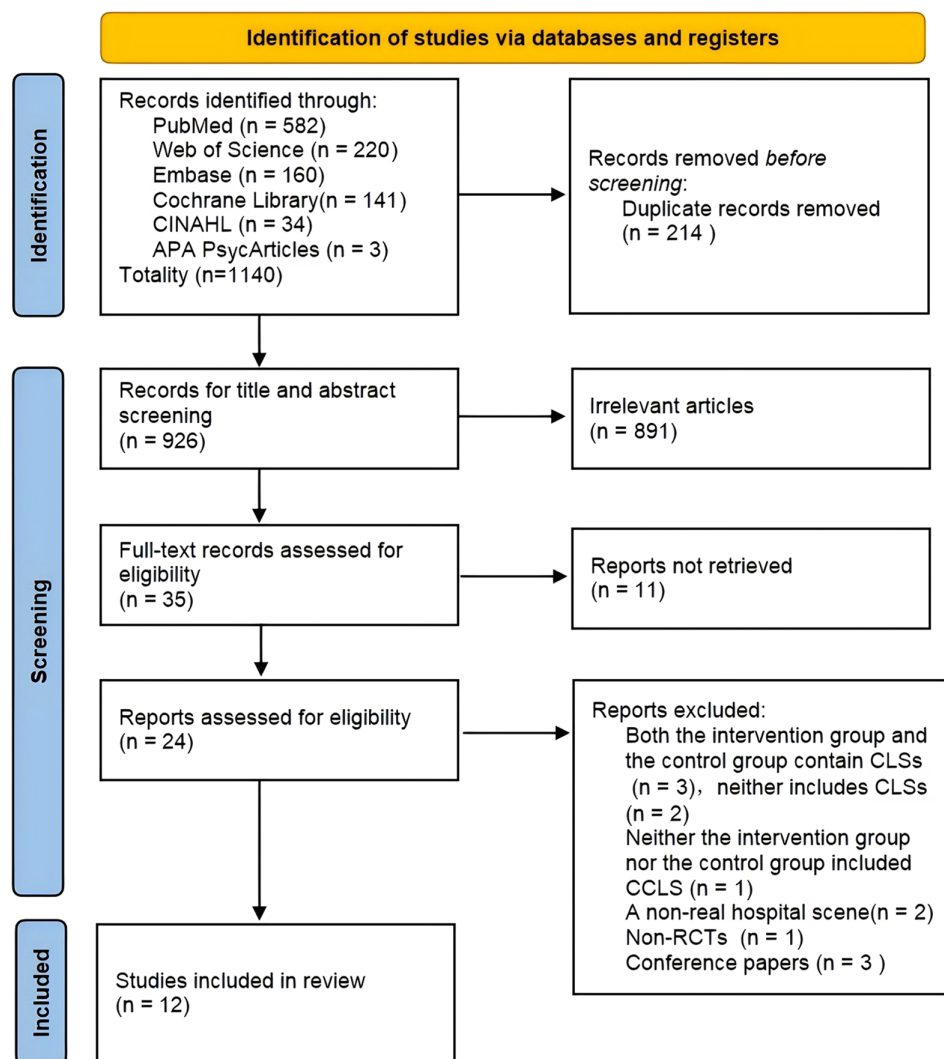


FIGURE 1 | Literature screening flowchart. *Note:* The flowchart illustrates the number of records identified from six databases (PubMed, Web of Science, Embase, Cochrane Library, CINAHL and APA PsycArticles), the number of records screened and excluded and the final number of studies included in the systematic review and meta-analysis ($n = 12$).

TABLE 1 | Basic characteristics of the included studies.

Studies	Country	Setting	Patients	Age bracket (years)	Sample size Total (EG1:EG2:CG)	Mean age			Intervenor
						EG1	EG2	CG	
Fletcher, S. 2023	Canada	Radiology department	Children requiring MRI	3–7	122 (39:36:47)	(5.6 ± 1.2) years	(5.5 ± 1.1) years	(5.6 ± 1.3) years	CCLS
Lee-Krueger, R. C. W. 2021	Canada	Ward	Children requiring venipuncture	4–12	103 (45:58)		(7.47 ± 2.67) years	(7.63 ± 2.56) years	Robot
Hinze, T. 2020	United States	Community dental clinics	Uncooperative paediatric dental patients	4–8	30 (14:16)		N	N	CCLS
West, N. 2020	Canada	Anaesthetic care unit	Children requiring venipuncture	3–10	59 (28:31)		5.5 (4, 7) years	4 (3, 7) years	CCLS
Diener, M. L. 2019	United States	The whole hospital	Children requiring venipuncture	> 2	95 (45: 50)		(114.20 ± 57.80) months	(115.56 ± 66.21) months	CCLS
Dumoulin, S. 2019	Canada	ED	Children requiring venipuncture	8–17	59 (24:20:15)	(12.65 ± 3.33) years	(13.85 ± 2.80) years	(13.81 ± 2.43) years	CCLS
Schlechter, J. A. 2017	United States	Paediatric orthopaedic cast room	Children with fractures	2–10	86 (42:44)		Mean 5 years and 9 months	Mean 6 years and 1 month	CCLS
Hyland, E. J. 2015	Australia	The burn unit	Children undergoing burn dressing changes	< 16	100 (50:50)		2.3 (1.5, 4.5) years	2.2 (1.6, 3.9) years	CLTP
Heilbrunn, B. R. 2014	United States	ED	Children in nonurgent and not requiring invasive procedures	5–12	113 (41:43:29)	(7.41 ± 2.09) years	(7.70 ± 2.26) years	(7.48 ± 1.96) years	CCLS
Tyson, M. E. 2014	United States	Radiology department	Children requiring imaging examinations	1–12	137 (68:69)		Mean 5 years old	Mean 4.5 years old	CCLS
Stevenson, M. D. 2005	United States	ED	Children requiring venipuncture	2–16	149 (76:73)		(8.1 ± 4.3) years	(8.8 ± 4.5) years	CCLS
Cavender, K. 2004	United States	ED	Children requiring venipuncture	4–11	43 (20:23)		(8.02 ± 1.58) years	(7.77 ± 1.89) years	CCLS

Note: This table summarizes the basic information of the 12 RCTs included in the analysis, including authors, country, setting, patient type, age range, sample size (total and group sizes), mean age and intervention provider (e.g., CCLS, CLTP or robot). 'EG' indicates experimental group(s), and 'CG' indicates control group. Age data are presented as mean ± standard deviation or median (interquartile range). Abbreviations: CCLS: certified child life specialist; CG: control group, CLTP: child life therapist; ED: emergency department; EG: experimental group; MRI: magnetic resonance imaging.

TABLE 2 | Intervention methods and primary outcome measures of the included studies.

Studies	Intervention			Timing of child life intervention	Registration	Informed consent	Primary outcomes
	EG1	EG2	CG				
Fletcher, S. 2023	Training with a child life specialist (no mock MRI)	Training in a mock MRI with a child life specialist	Home-based preparation materials	Training occurred a few days prior to their MRI	Y	Y	PedsQL VAS
Lee-Krueger, R. C. W. 2021		A humanoid robot (MEDiVR)—pre- programmed based on child life specialist input	Standard care	Mean 42.5-min pre-procedure	Y	Y	① FPS-R ② CFS
Hinze, T. 2020		Child life services	Standard care	One-week and 1-day pretreatment	Y	Y	① Houpt scale
West, N. 2020		Child life services	Standard care	Only during the preoperative period on the day of surgery	Y	Y	① mYPAS-SF ② PACBIS
Diener, M. L. 2019		Child life services	Standard care	Pre-puncture	Y	Y	① OSBD
Dumoulin, S. 2019	Watching television	VR	Child life services	During puncture	Y	Y	① VAS ② Children's and Parents' Satisfaction
Schlechter, J. A. 2017		Child life services	Standard care	During the procedure	Y	Y	① Heart rate ② Subjective behaviour score
Hyland, E. J. 2015		Child life services	Standard care	Before and during the procedure	Y	Y	① CHEOPS ② Children's Fear Scale
Heilbrunn, B. R. 2014	Child life intervention	Clown intervention	Standard care	Before the procedure	Y	Y	mYPAS
Tyson, M. E. 2014		Child life services	Standard care	During the examinations	Y	Y	① Parent Perceptions of Specialty Care ② Children's perception of pain and distress
Stevenson, M. D. 2005		Child life services	Standard care	During puncture	Y	Y	① OSBD-r ② State-Trait Anxiety Inventory for Children

(Continues)

TABLE 2 | (Continued)

Studies	Intervention			Timing of child life intervention	Registration	Informed consent	Primary outcomes
	EG1	EG2	CG				
Cavender, K. 2004		Child life services	Standard care	During puncture	Y	Y	① The FACES scale ② The Glasses Fear scale ③ PBCL

Note: This table details the specific interventions for each experimental group (EG1, EG2) and control group (CG), the timing of child life intervention, whether the trial was registered, whether informed consent was obtained and the primary outcome measures in each included study.

Abbreviations: CFS: Child Fear Scale; C-Ped-PROMIS: Chinese Version of Paediatric Patient-Reported Outcomes Measurement Information System; FPS-R: Faces Pain Scale-Revised; mYPAS-SF: Modified Yale Preoperative Anxiety Scale—Short Form; OSBD: Observational Scale of Behavioural Distress; OSBD-r: Observation Scale of Behaviour Distress-revised; PACBIS: Perioperative Adult Child Behaviour Interaction Scale; PBCL: Procedural Behaviour Checklist; PedsQL VAS: Peds QL Present Functioning Visual Analogue Scales; SDSC: Sleep Disturbance Scale for Children; VR: virtual reality.

Dumoulin et al. 2019; Heilbrunn et al. 2014; Tyson et al. 2014; Cavender et al. 2004) provided allocation concealment schemes, and five studies (Lee-Krueger et al. 2021; Stevenson et al. 2005; Hinze et al. 2020; West et al. 2020; Heilbrunn et al. 2014) explicitly mentioned blinding of outcome assessors. The quality of the included studies was moderate to high. The risk-of-bias assessment is presented in Figure 2.

3.1 | Meta-Analysis

The summary of meta-analysis results is presented in Table 3, and the forest plots for all outcome indicators are shown in Figures 3–6.

3.1.1 | Pain

Five RCTs (Lee-Krueger et al. 2021; Fletcher et al. 2023; Dumoulin et al. 2019; Hyland et al. 2015; Cavender et al. 2004) ($n=343$) reported pain outcomes. Overall, CLS interventions significantly reduced paediatric pain compared to controls ($p=0.009$; Figure 3). Sensitivity analysis excluding a robot-assisted intervention study (Lee-Krueger et al. 2021) confirmed the stability and robustness of this treatment effect.

3.1.2 | Fear

Five studies (Lee-Krueger et al. 2021; Fletcher et al. 2023; Dumoulin et al. 2019; Hyland et al. 2015; Cavender et al. 2004) ($n=356$) reported children's fear. Pooled estimates showed no statistically significant reduction in fear following CLS interventions ($p=0.290$; Figure 4), a finding that remained consistent in sensitivity analysis after excluding the robot-assisted trial (Lee-Krueger et al. 2021).

3.1.3 | Distress

The outcome measures of three studies (Stevenson et al. 2005; Fletcher et al. 2023; Diener et al. 2019) ($n=330$) included behavioural distress in paediatric patients. Pooled analysis revealed that CLS interventions significantly reduced paediatric behavioural distress compared to standard care ($p=0.018$; Figure 5).

3.1.4 | Anxiety

Four studies (Stevenson et al. 2005; Fletcher et al. 2023; West et al. 2020; Heilbrunn et al. 2014) reported on child anxiety. Among these, the study by Stevenson et al. (2005) was excluded from the meta-analysis because it exclusively enrolled children aged ≥ 8 years and lacked exact group sample sizes for data extraction; qualitatively, it reported no significant effect of CLS services during emergency venipuncture. Consequently, three studies (Fletcher et al. 2023; West et al. 2020; Heilbrunn et al. 2014) ($n=215$) were quantitatively synthesized. Pooled estimates demonstrated that CLS interventions significantly reduced paediatric anxiety ($p=0.007$; Figure 6).

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Cavender, K. 2004	+	+	-	?	+	+	+
Diener, M. L. 2019	-	?	?	?	+	+	+
Dumoulin, S. 2019	+	+	-	?	+	+	+
Fletcher, S. 2023	+	-	-	?	+	+	+
Heilbrunn, B. R. 2014	+	+	-	+	+	+	+
Hinze, T. 2020	+	?	-	+	+	+	+
Hyland, E. J. 2015	+	?	-	?	+	+	+
Lee-Krueger, R. C. W. 2021	+	+	-	+	+	+	+
Schlechter, J. A. 2017	+	?	-	?	+	+	+
Stevenson, M. D. 2005	+	+	-	+	+	+	+
Tyson, M. E. 2014	+	+	-	?	+	+	+
West, N. 2020	+	+	-	+	+	+	+

FIGURE 2 | Risk-of-bias assessment results for the included studies. *Note:* The figure presents the methodological quality of the 12 randomized controlled trials evaluated using the Cochrane Risk of Bias tool (version 5.1.0). Green circles indicate low risk of bias, yellow circles indicate unclear risk of bias, and red circles indicate high risk of bias for each domain.

3.1.5 | Other Outcomes Assessed

Regarding patient compliance, Hinze et al. (2020) used the modified Houtp scale and found that CLSs did not significantly improve compliance in children undergoing dental treatment compared to standard care. Schlechter et al. (2017) evaluated the effect of CLSs on children using physiological data—heart rate—and showed that CLSs significantly reduced heart rate in children with fractures during treatment.

4 | Discussion

The results of this review indicate that the involvement of CLSs in healthcare settings can reduce pain, psychological distress

and anxiety in paediatric patients during nursing procedures and clinical treatments. However, the effect on alleviating children's fear was not significant. These findings affirm the potential value of CLSs in paediatric psychological care while also highlighting areas requiring further investigation.

A previous systematic review by Hummel et al. (2023) reached conclusions consistent with ours, despite differences in the studies included, both affirming the positive role of CLSs on children's mental health in clinical settings. The present review extends and deepens previous research in several ways. First, by expanding database searches beyond those by Hummel et al. (2023) (adding Web of Science, Cochrane Library and CINAHL), this review identified nine new eligible RCTs. We retained three RCTs (Schlechter et al. 2017; Hyland et al. 2015; Tyson et al. 2014) from Hummel et al. (2023), while excluding one community-based study that violated our strict clinical setting criteria. Second, regarding outcome measures, this study additionally analysed anxiety and distress, providing a more comprehensive assessment of the effects of CLS intervention. Third, this study broadened the inclusion criteria—the intervention group was required to involve CLS participation, either directly or indirectly. Therefore, this review is the first to include studies related to humanoid robots (MEDi) (Lee-Krueger et al. 2021) programmed based on the expertise of CLSs.

Although the study by Lee-Krueger et al. (2021) utilized MEDi instead of CLSs for intervention, all actions were pre-programmed based on the best practice recommendations of CLSs and presented uniformly to all subjects. We regard MEDi merely as a medium for information delivery rather than as an independent agent; thus, it was included in this study. Sensitivity analyses exclude the study by Lee-Krueger et al. (2021), which demonstrated that the overall conclusions remained unchanged, with no significant alterations in heterogeneity or effect sizes. However, it is important to acknowledge that this approach differs from direct intervention by CLSs. The findings of the study (Lee-Krueger et al. 2021) indicate that child life services provided by MEDi are not superior to standard care in alleviating pain and fear in paediatric patients. This may be primarily due to the lack of personalization in pre-programming, which cannot provide adaptive responses based on the child's age, personality, emotional state, family dynamics and environmental factors. Nevertheless, in an era where digital and artificial intelligence technologies are rapidly integrating into the medical field, this presents a new avenue for clinicians and researchers to explore. For instance, combining the expertise and skills of CLSs with artificial intelligence could enable personalized responses and countermeasures during interactions with patients, thereby better meeting clinical needs. Given the limited research utilizing robots as a medium, further rigorous research designs are necessary to investigate whether this approach can complement traditional methods in this field.

In medical settings, children's fear often stems from the unknown or prior negative experiences. This review found no significant effect of CLS interventions on fear. First, three (Lee-Krueger et al. 2021; Dumoulin et al. 2019; Cavender et al. 2004) of the five (Lee-Krueger et al. 2021; Fletcher et al. 2023;

TABLE 3 | Summarized results of meta-analyses.

Outcome	Included studies	Sample size	Effect size (SMD)	95% CI	p	Heterogeneity (I^2)
Pain						
Overall analysis (all studies)	5	343	-0.28	[-0.49, -0.07]	0.009	0%
Human-delivered CLS only	4	240	-0.37	[-0.62, -0.13]	0.003	0%
Fear						
Overall analysis (all studies)	5	356	-0.11	[-0.32, 0.10]	0.29	0%
Human-delivered CLS only	4	253	-0.22	[-0.47, -0.02]	0.073	0%
Distress	3	330	-0.26	[-0.48, -0.05]	0.018	0%
Anxiety	3	215	-0.37	[-0.65, -0.01]	0.007	0%

Note: This table summarizes the primary quantitative syntheses across paediatric pain, fear, distress and anxiety outcomes. Data are presented as pooled effect sizes, SMD, 95% CI, p-values and I^2 .

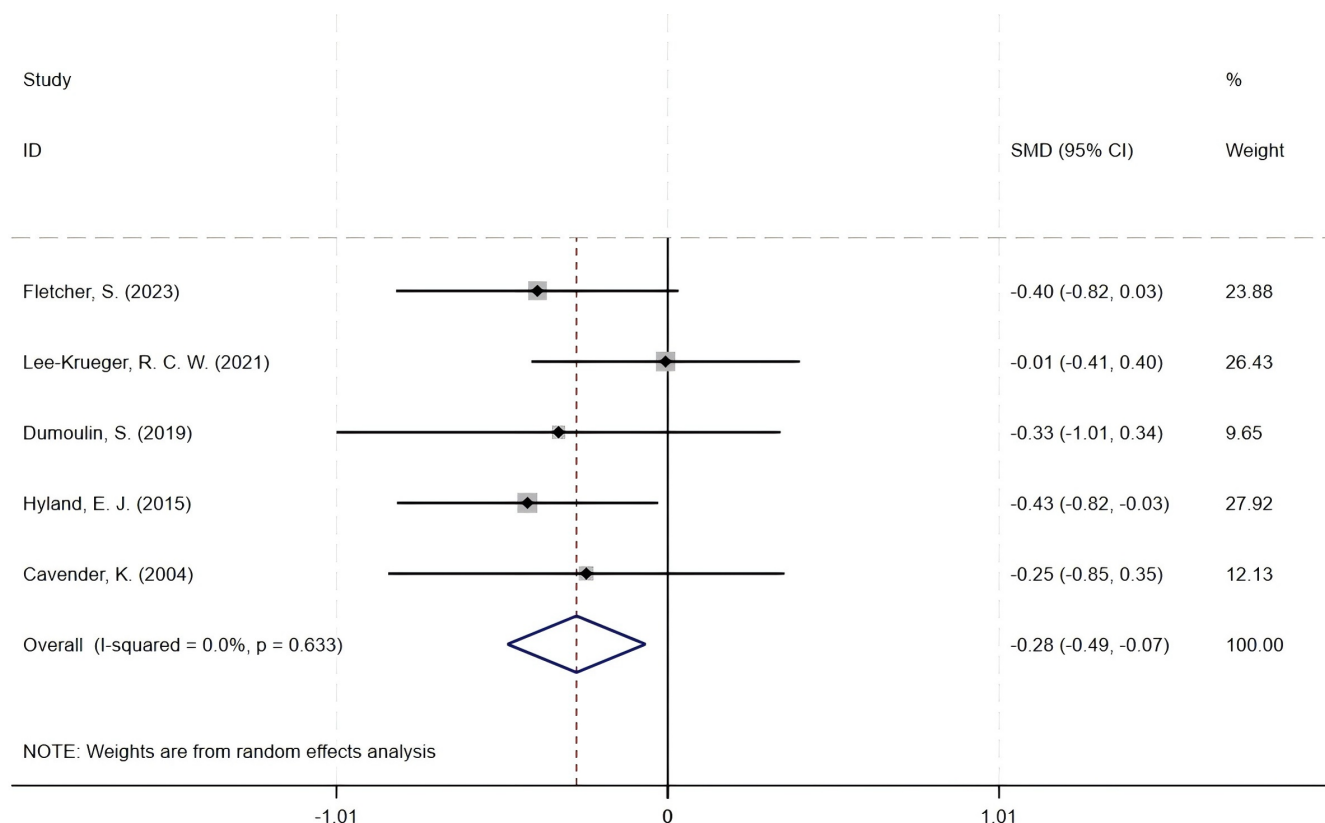


FIGURE 3 | Forest plot of the effect of CLSs on children's pain. Note: Forest plot of the effect of CLS intervention on children's pain. Squares represent the SMD of each study, with the size proportional to the weight in the meta-analysis. Horizontal lines indicate 95% CI. The diamond represents the pooled effect size calculated using a random-effects model (SMD = -0.28, 95% CI -0.49 to -0.07, $p = 0.009$). Heterogeneity was low ($I^2 = 0\%$, $p = 0.633$).

Dumoulin et al. 2019; Hyland et al. 2015; Cavender et al. 2004) included studies involved venipuncture. Needle fear typically arises from repeated pain-needle pairings, leading to the development of conditioned fear (Field 2006). Although CLS strategies such as information, play and distraction may reduce

situational anxiety, they cannot eliminate established conditioned fear. Additionally, CLS interventions in some studies were limited to brief pre-procedural preparation or immediate intra-procedural distraction. For highly fearful children, a single short-term intervention is insufficient to modify entrenched

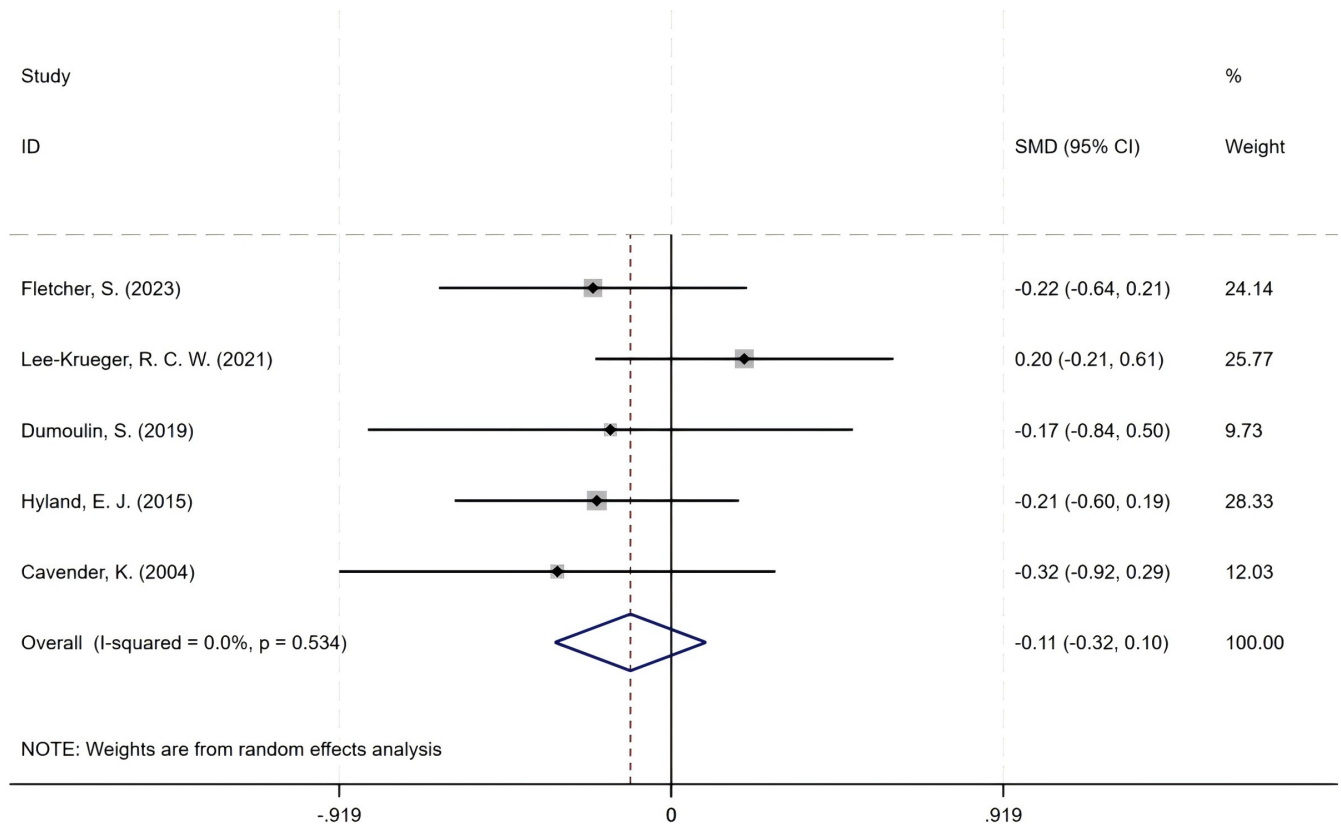


FIGURE 4 | Forest plot of the effect of CLSs on children's fear. *Note:* Forest plot of the effect of CLS intervention on children's fear. Squares represent the SMD of each study, with the size proportional to the weight in the meta-analysis. Horizontal lines indicate 95% CI. The diamond represents the pooled effect size calculated using a random-effects model (SMD = -0.11, 95% CI -0.32 to 0.10, $p = 0.290$). Heterogeneity was low ($I^2 = 0\%$, $p = 0.534$).

fear patterns, which is consistent with evidence suggesting that multiple exposure therapy sessions outperform a single session (McMurtry et al. 2015). Finally, physical restraint during invasive procedures may intensify fear, underscoring the necessity for trauma-informed care (Goddard et al. 2022).

Accurately measuring paediatric subjective experiences remains a significant challenge. First, the studies included in this review exhibited substantial heterogeneity in measurement tools, particularly concerning subjective internalizing constructs such as fear and anxiety. This variability likely amplified measurement errors across the trials. Second, although this review strictly synthesized child self-reports for methodological consistency, ratings from younger children are susceptible to potential reporting bias, stemming from limitations in cognitive development, linguistic skills and medical understanding. Third, proxy reports (e.g., parent- or clinician-rated scales) were not universally available across trials, restricting our capacity to evaluate inter-rater concordance. Future research should emphasize tool standardization and adopt a multimodal evaluation framework (Liu et al. 2025). This framework should integrate child self-reports, proxy ratings, objective physiological markers (e.g., heart rate, Schlechter et al. 2017; salivary biomarkers, Fortunato C et al. 2008), behavioural metrics (e.g., adherence and willingness for future care) and health economic endpoints (e.g., length of stay and total medical costs).

The content of CLS interventions, such as explanation and preparation, medical play, distraction and family support,

presents a common challenge in behavioural intervention research: Neither the participants (children and their parents) nor the CLSs can be blinded to group allocation. Consequently, the trials included in this study are subject to open-label limitations concerning allocation blinding, which may introduce a positive bias in the conclusions regarding the effectiveness of CLS interventions on children's pain, psychological distress and anxiety. Specifically, children and parents, upon being informed that they have received the 'extra care' intervention, may actively alter their behaviour or tend to report lower levels of pain or fear, thereby exaggerating the true effect of subjective outcomes. To address this, future trials should implement strict hypothesis blinding for participants and standardize control group experiences (e.g., accounting for usual-care placebo effects) to minimize subjective bias. Among the 12 RCTs reviewed, only five explicitly mentioned blinding of outcome assessors, further potentially compromising the validity of outcome evaluation. It is recommended that future CLS-related trials enhance the methodological rigour of study design by routinely implementing blinding of outcome assessors and ensuring that assessors remain unaware of both the specific study hypotheses and group allocation to minimize detection and performance bias.

Among the included studies, CLSs employ a comprehensive range of intervention strategies, including explanation and preparation, medical play, distraction and family support. Notably, distraction techniques—such as conversing,

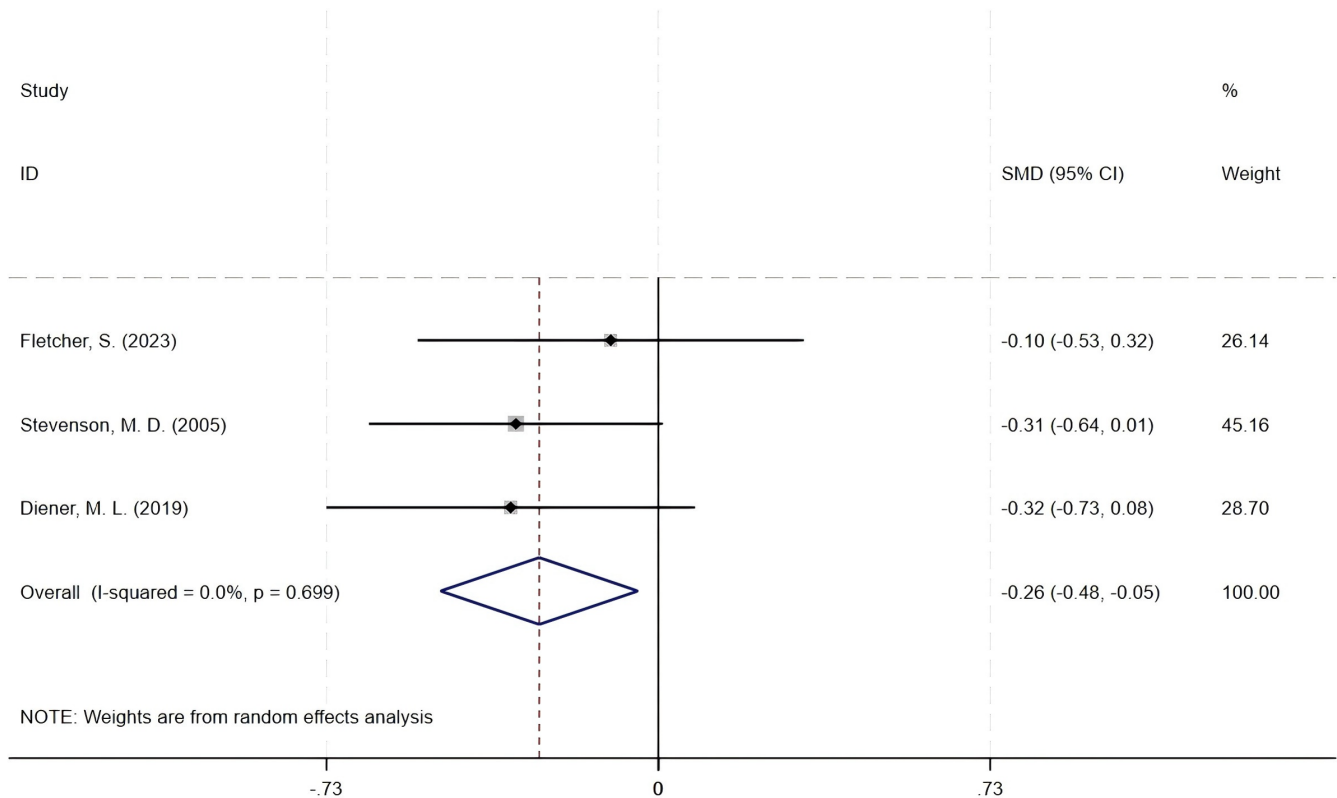


FIGURE 5 | Forest plot of the effect of CLSs on children's distress. *Note:* Forest plot of the effect of CLS intervention on children's distress. Squares represent the *SMD* of each study, with the size proportional to the weight in the meta-analysis. Horizontal lines indicate 95% *CI*. The diamond represents the pooled effect size calculated using a random-effects model (*SMD* = -0.26, 95% *CI* -0.48 to -0.05, *p* = 0.018). Heterogeneity was low (*I*² = 0%, *p* = 0.699).

reading books or engaging with toys—have proven effective in alleviating children's negative emotions in certain instances (Schlechter et al. 2017; Dumoulin et al. 2019; Cavender et al. 2004). Furthermore, Fletcher et al. (2023) demonstrated that even basic home-based preparation materials, such as picture books, videos and audio recordings, yielded a 96% efficacy rate in unsedated MRI scans, serving as a valuable reference for clinical practitioners in settings lacking a dedicated CLS. Additionally, interventions should be customized according to the child's developmental stage: For infants and toddlers, parental presence and sensory soothing are essential; for preschoolers, the use of concrete language and correction of misconceptions regarding 'punishment' is crucial; for school-aged children, visual aids and the provision of choices are beneficial; for adolescents, respecting autonomy and privacy is important; and for children with developmental delays, individualized environmental management and family collaboration are necessary (Panella 2016). Finally, it is imperative to consider parental satisfaction and the completion rates of various treatments. The presence of CLS can reduce the sedation requirements for paediatric patients during MRI examinations (Fletcher et al. 2023; Scott et al. 2016) and enhance satisfaction among both staff and parents (Tyson et al. 2014). These advantages collectively demonstrate the cost-effectiveness of CLS services.

For parents, the following four methods can be beneficial for affected children. First, it is beneficial to prepare in advance by utilizing picture books, short videos or audio recordings

that depict the hospital setting (Fletcher et al. 2023; West et al. 2020). Second, actively engaging the child in distraction techniques during procedures, such as games like 'I spy' or using bubbles, proves to be more effective than providing passive reassurance. Third, offering physical comfort by holding the child upright on the lap, as opposed to lying them flat, can significantly reduce fear and the necessity for restraint (Cavender et al. 2004). Finally, maintaining a calm demeanour is crucial, as a parent's tranquillity positively impacts the child's emotional state, especially in younger children (Panella 2016).

4.1 | Limitations and Prospects

Most outcome measures attained statistical significance in the current study; however, there are limitations. Firstly, we searched six major databases, and then we included only 12 RCTs. Participants in meta-analyses were relatively small. Additionally, data transformation methods (Wan et al. 2014) were used for studies by Fletcher et al. (2023), Hyland et al. (2015), and Stevenson et al. (2005), which may have affected the statistical power of the results. Secondly, all 12 studies included in this review were conducted in Western countries such as the United States, Canada and Australia. As a consequence of variations in care settings, healthcare systems and cultural attributes, any application of this research's findings to non-Western cultures should be approached with caution. Future studies should examine whether CLS interventions are applicable and culturally

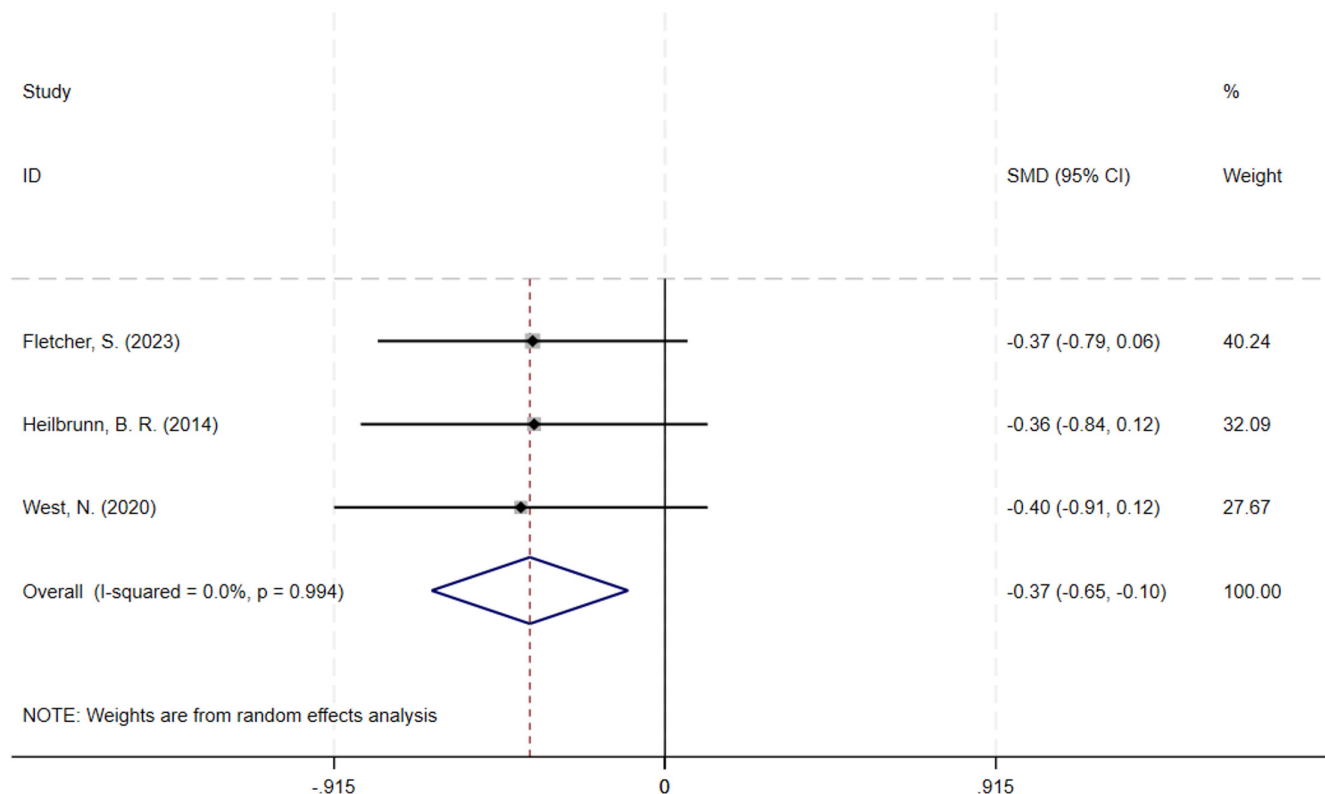


FIGURE 6 | Forest plot of the effect of CLSs on children's anxiety. *Note:* Forest plot of the effect of CLS intervention on children's anxiety. Squares represent the *SMD* of each study, with the size proportional to the weight in the meta-analysis. Horizontal lines indicate 95% *CI*. The diamond represents the pooled effect size calculated using a random-effects model (*SMD* = -0.37 , 95% *CI* -0.65 to -0.01 , $p = 0.007$). Heterogeneity was low ($I^2 = 0\%$, $p = 0.994$).

adaptable to children from various non-Western background. Lastly, no studies included in this review reported long-term consequences of CLS interventions. As early childhood medical experiences can play a critical role in influencing future behaviours towards medical touch, future actions could introduce long-term follow-up.

5 | Conclusion

In summary, the findings of this systematic review indicate that CLS interventions demonstrate positive potential in improving the relief of pain, anxiety and psychological distress in children during medical care. Although the effect of CLSs on alleviating paediatric fear remains inconclusive, their potential value in clinical practice warrants broader recognition and deeper integration into paediatric care. In addition to supplementing the body of evidence with high-quality RCTs to refine study conclusions, future research holds significant promise in exploring digital technologies to expand the accessibility of CLS services.

Author Contributions

Fangqin Li and Jianli Jin: Conception and design, literature search, data extraction, quality assessment and drafting of the initial manuscript. Huanhuan Li and Liwei Feng: Statistical analysis and forest plot generation. Fangqin Li: Manuscript revision. Wenjiao Huang: Critical revision

of the manuscript for important intellectual content. All authors read and approved the final manuscript.

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Ethics Statement

The authors have nothing to report.

Consent

Patient consent was not required because this study is a systematic review of previously published literature and does not contain any individual person's data.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data used in this systematic review are derived from previously published studies cited in the reference list. No new primary data were generated or analysed.

References

- Abdi, F., A. Karamoozian, M. Lotfilou, et al. 2025. "Effect of Play Therapy and Storytelling on the Anxiety Level of Hospitalized Children: A Randomized Controlled Trial." *BMC Complementary Medicine and Therapies* 25, no. 1: 23.
- Ali, S., R. Manaloor, K. Ma, et al. 2021. "A Randomized Trial of Robot-Based Distraction to Reduce Children's Distress and Pain During Intravenous Insertion in the Emergency Department." *CJEM* 23, no. 1: 85–93.
- Baghele, A., N. Dave, R. Dias, and H. Shah. 2019. "Effect of Preoperative Education on Anxiety in Children Undergoing Day-Care Surgery." *Indian Journal of Anaesthesia* 63, no. 7: 565–570.
- Baliga, S., R. Ravikiran S, R. Rao, and S. P. Kamath. 2025. "Premedical Procedure Anxiety for Intravenous Cannulation in Hospitalized Children: A Facility-Based Cross-Sectional Study." *International Journal of Clinical Pediatric Dentistry* 18, no. 11: 1333–1338.
- Bedner, A., D. Gonzalez, S. Hayward, et al. 2021. "232 A Randomized Controlled Trial on Virtual Reality Use as a Distraction Technique for Orthopedic Injuries Requiring Sedation." *Annals of Emergency Medicine* 78, no. 4: S94.
- Brown, S., P. Cordner, K. A. Weir, and T. Machingura. 2022. "A Systematic Scoping Literature Review on the Use of Child Life Therapy (CLT) in the Perioperative Setting." *Journal of Perioperative Practice* 32, no. 10: 244–259.
- Bryl, A. W., B. Bonsu, L. Johnson A, K. B. J. Pommert, K. A. Hollenbach, and J. T. Kanegaye. 2021. "Tablet Computer as a Distraction Tool During Facial Laceration Repair: A Randomized Trial." *Pediatric Emergency Care* 37, no. 8: e425–e430.
- Cavender, K., M. D. Goff, C. Hollon E, and C. E. Guzzetta. 2004. "Parents' Positioning and Distracting Children During Venipuncture. Effects on Children's Pain, Fear, and Distress." *Journal of Holistic Nursing* 22, no. 1: 32–56.
- Dehghan, Z., T. Reyhani, V. Mohammadpour, S. Z. Aemmi, R. Shojaeian, and S. M. Asghari Nekah. 2017. "The Effectiveness of Dramatic Puppet and Therapeutic Play in Anxiety Reduction in Children Undergoing Surgery: A Randomized Clinical Trial." *Iranian Red Crescent Medical Journal* 19, no. 3: 1–8.
- Diener, M. L., O. Lofgren A, A. Isabella R, S. Magana, C. Choi, and C. Gourley. 2019. "Children's Distress During Intravenous Placement: The Role of Child Life Specialists." *Children's Health Care* 48, no. 1: 103–119.
- Doupnik S, K., J. Lawlor, B. T. Zima, et al. 2018. "Mental Health Conditions and Unplanned Hospital Readmissions in Children." *Journal of Hospital Medicine* 13, no. 7: 445–452.
- Dumoulin, S., S. Bouchard, J. Ellis, et al. 2019. "A Randomized Controlled Trial on the Use of Virtual Reality for Needle-Related Procedures in Children and Adolescents in the Emergency Department." *Games for Health Journal* 8, no. 4: 285–293.
- Field, A. P. 2006. "Is Conditioning a Useful Framework for Understanding the Development and Treatment of Phobias?" *Clinical Psychology Review* 26, no. 7: 857–875.
- Fletcher, S., D. Lardner, M. Bagshawe, et al. 2023. "Effectiveness of Training Before Unsedated MRI Scans in Young Children: A Randomized Control Trial." *Pediatric Radiology* 53, no. 7: 1476–1484.
- Fletcher, S., D. Lardner, M. Bagshawe, et al. 2021. "Evaluating Preparation Methods for Clinical MRI Scans in Young Children Without Anesthesia." *Anesthesia and Analgesia* 133, no. 3 SUPPL 2: 1252–1253.
- Fortunato C, K., E. Dribin A, A. Granger D, and K. A. Buss. 2008. "Salivary Alpha-Amylase and Cortisol in Toddlers: Differential Relations to Affective Behavior." *Developmental Psychobiology* 50, no. 8: 807–818.
- Goddard, A., E. Janicek, and L. Etcher. 2022. "Trauma-Informed Care for the Pediatric Nurse." *Journal of Pediatric Nursing* 62: 1–9.
- Goldman R, D., and A. Behboudi. 2021. "Virtual Reality for Intravenous Placement in the Emergency Department—A Randomized Controlled Trial." *European Journal of Pediatrics* 180, no. 3: 725–731.
- Green, S., and J. P. T. Higgins. 2011. "Cochrane Handbook for Systematic Reviews of Interventions Version 5.1.0[R]." The Cochrane Collaboration.
- Heilbrunn, B. R., E. Wittern R, J. B. Lee, P. K. Pham, A. H. Hamilton, and A. L. Nager. 2014. "Reducing Anxiety in the Pediatric Emergency Department: A Comparative Trial." *Journal of Emergency Medicine* 47, no. 6: 623–631.
- Higgins, J. P. T., G. Thompson S, J. Deeks J, J. P. Higgins, and D. G. Altman. 2003. "Measuring Inconsistency in Meta-Analyses." *British Medical Journal (Clinical Research Edition)* 327, no. 7414: 557–560.
- Hinze, T., C. McDonald, C. A. Kerins, et al. 2020. "Child Life Interventions for Pediatric Dental Patients: A Pilot Study." *Pediatric Dentistry* 42, no. 4: 252–255.
- Hummel, J., M. Coenen, V. Voigt-Blaurock, C. Klein, and C. Jung-Sievers. 2023. "'Child Life Specialist' Interventions in Clinical Pediatric Care: A Systematic Review of the Effects on Mental Health Outcomes of Children and Adolescents." *Das Gesundheitswesen* 85, no. 1: 39–47.
- Hyland, E. J., R. D'Cruz, G. Harvey J, J. Moir, C. Parkinson, and A. J. A. Holland. 2015. "An Assessment of Early Child Life Therapy Pain and Anxiety Management: A Prospective Randomised Controlled Trial." *Burns* 41, no. 8: 1642–1652.
- Lee-Krueger, R. C. W., R. Pearson J, A. Spencer, M. Noel, L. Bell-Graham, and T. N. Beran. 2021. "Children's Pain During IV Induction: A Randomized-Controlled Trial With the MEDi Robot." *Journal of Pediatric Psychology* 46, no. 8: 991–1000.
- Liu, J., T. Li, M. Zhu, Q. Lin, G. Jing, and D. Jinlan. 2025. "Measurement Properties of Instruments to Assess Medical Fear in Children: A Systematic Review Protocol." *BMJ Open* 15, no. 11: e101603.
- McMurtry, C. M., M. Noel, A. Taddio, et al. 2015. "Interventions for Individuals With High Levels of Needle Fear: Systematic Review of Randomized Controlled Trials and Quasi-Randomized Controlled Trials." *Clinical Journal of Pain* 31, no. 10 Suppl: S109–S123.
- Mikuluk, B., P. Guttoo, A. Anderson, and M. Skeens. 2023. "Certified Child Life Specialist Role in Implementation of Individualized Coping Plans for Children Receiving Botulinum Toxin Injections." *Journal for Specialists in Pediatric Nursing* 28, no. 3: e12413.
- Moura, L. A. D., I. M. G. Dias, and V. Pereira L. 2016. "Prevalence and Factors Associated With Preoperative Anxiety in Children Aged 5–12 Years." *Revista Latino-Americana de Enfermagem* 24: e2708.
- Page, M. J., E. McKenzie J, M. Bossuyt P, et al. 2020. "The PRISMA 2020 Statement: An Updated Guideline for Reporting Systematic Reviews." *Journal of Clinical Epidemiology* 134: 178–189.
- Panella, J. J. 2016. "Preoperative Care of Children: Strategies From a Child Life Perspective." *AORN Journal* 104, no. 1: 11–22.
- Romito, B., J. Jewell, M. Jackson, et al. 2021. "Child Life Services." *Pediatrics* 147, no. 1: e2020040261.
- Schlechter, J. A., A. L. Avik, and S. DeMello. 2017. "Is There a Role for a Child Life Specialist During Orthopedic Cast Room Procedures? A Prospective-Randomized Assessment." *Journal of Pediatric Orthopaedics. Part B* 26, no. 6: 575–579.
- Scott, M. T., K. E. Todd, H. Oakley, et al. 2016. "Reducing Anesthesia and Health Care Cost Through Utilization of Child Life Specialists in Pediatric Radiation Oncology." *International Journal of Radiation Oncology, Biology, Physics* 96, no. 2: 401–405.
- Stevenson, M. D., M. Bivins C, K. O'Brien, and J. A. Gonzalez del Rey. 2005. "Child Life Intervention During Angiocatheter Insertion in the

Pediatric Emergency Department.” *Pediatric Emergency Care* 21, no. 11: 712–718.

Tyson, M. E., D. D. Bohl, and J. G. A. Blickman. 2014. “A Randomized Controlled Trial: Child Life Services in Pediatric Imaging.” *Pediatric Radiology* 44, no. 11: 1426–1432.

Wan, X., W. Wang, J. Liu, and T. Tong. 2014. “Estimating the Sample Mean and Standard Deviation From the Sample Size, Median, Range and/or Interquartile Range.” *BMC Medical Research Methodology* 14: 135.

West, N., N. Christopher, K. Stratton, M. Görges, and Z. Brown. 2020. “Reducing Preoperative Anxiety With Child Life Preparation Prior to Intravenous Induction of Anesthesia: A Randomized Controlled Trial.” *Paediatric Anaesthesia* 30, no. 2: 168–180.

Wittenberg B, M., and D. Barnhart. 2025. “How Are Certified Child Life Specialists Perceived by Healthcare Professionals? A Call for Interprofessional Collaboration.” *Journal of Interprofessional Care* 39, no. 6: 924–932.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting information.